

THE MOST PROFITABLE

Trading Strategy Ever Documented

The Jim Simons / Medallion Fund system — explained in plain English,
with pictures, analogies, and lessons you can actually use.

THE 30-SECOND ANSWER: From 1988–2018, Renaissance Technologies' **Medallion Fund** turned **\$100** into **~\$398.7 million** — about **66% per year before fees (~39–40% after fees)** — and **never had a losing year**. No other documented strategy comes close over 30+ years. This guide explains **how it works, in simple terms**.

66% avg / year before fees	~40% avg / year after fees	0 losing years in 31 yrs	150k+ trades on big days
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Based on a deep web search: academic reconstructions, the book *The Man Who Solved the Market*, fund reporting & data studies •
September 2026

Inside: proof & numbers • how the strategy works • 7 secrets in simple words • a worked example • what YOU can copy • honest
warnings • 30-day plan

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How to read this guide: You don't need math. Every idea comes with a **real-life analogy** and a **"so what do I do?"** line. Skim the green boxes for the key lessons if you're in a hurry.

1 • The answer in 30 seconds — who wins?

After searching across academic papers, fund records, books and data studies, **one winner stands alone**:

The Renaissance Technologies Medallion Fund (Jim Simons' quant fund, launched 1988) is the **most profitable trading strategy ever documented** over a long period: roughly **66% a year before fees / ~39–40% after fees**, compounded for **30+ years with no losing year**. One academic reconstruction: **\$100 → \$398.7 million (1988–2018)**.

What IS the strategy, in simple words? Medallion is a **100% computer-run system** that scans hundreds of markets, finds **tiny, repeating price patterns** humans can't see, and trades them **thousands of times a day** — holding from minutes to a few days — while staying **hedged, diversified and strictly risk-controlled**, with **leverage** to turn small edges into huge returns. No gut feelings. No CNBC opinions. Just data + discipline + repetition.

Why it beats everything else: Warren Buffett (~20% for 60 years) is the greatest *investor*. Larry Williams (+11,376% in one year) had the greatest *single year*. But nobody else has documented **~40% net for 30+ years without a down year** — including straight through the dot-com crash and the 2008 crisis, when Medallion was *up* while the world was down.

2 • The proof: numbers that broke Wall Street's brain

Finance professor Bradford Cornell called Medallion "the ultimate counterexample" to the idea that markets can't be beaten — returns "far outstrip anything reported in the academic literature." Here are the headline facts:

Metric	Medallion Fund	What it means (simply)
Gross return 1988–2018	~63–66% / year	Before fees. The raw power of the system.
Net return (what investors kept)	~39–40% / year	After Medallion's giant 5% + 44% fees. Still ~4× the S&P.;
\$100 invested in 1988 → 2018	→ ~\$398,700,000 (gross)	Same \$100 in S&P; 500 → ~\$1,910.
Losing years (31 years)	Zero	Never down — even in 2000–02 and 2008.
2008 financial crisis	~+82% net (~+152% gross)	S&P; 500: -37%. Medallion <i>thrives</i> in chaos.
Fees charged	5% + 44% of profits	Double the normal 2-and-20. Still worth it.
Who can invest?	Employees only (since 1993)	You can't buy it. Fund capped ~\$10–15B.

Growth of \$100: Medallion Fund vs S&P 500 (1988-2018, log scale)

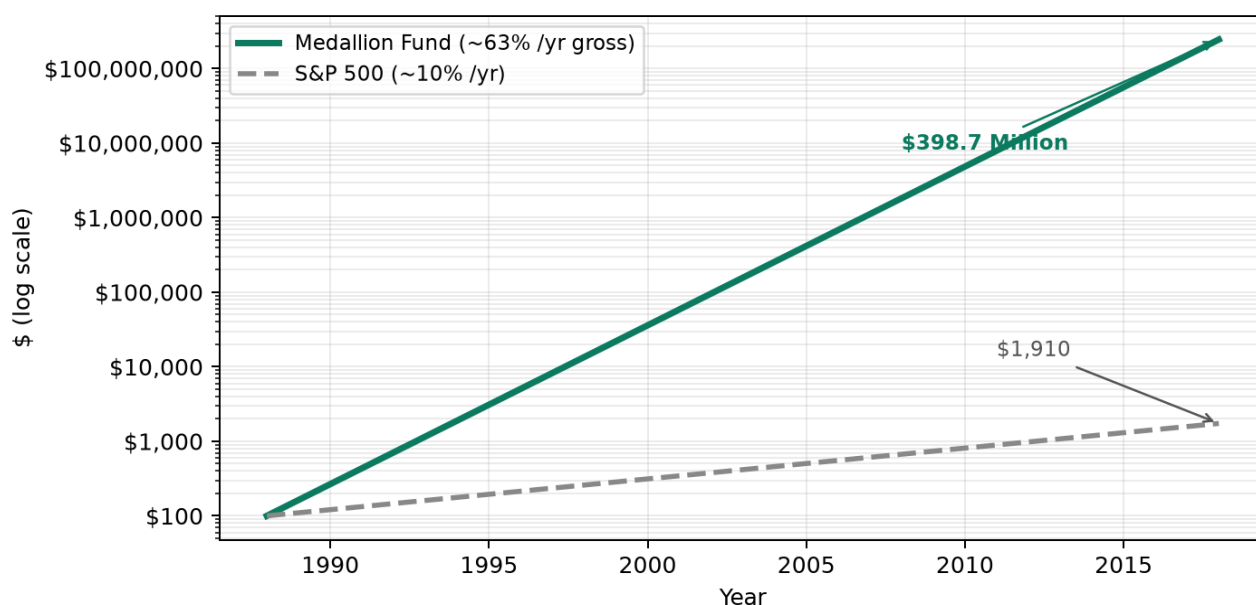


Figure 1 — \$100 growing at Medallion's ~63% vs the S&P's ~10% (log scale). By 2018 the gap is \$398.7M vs \$1,910. Based on Cornell's reconstruction of reported gross returns.

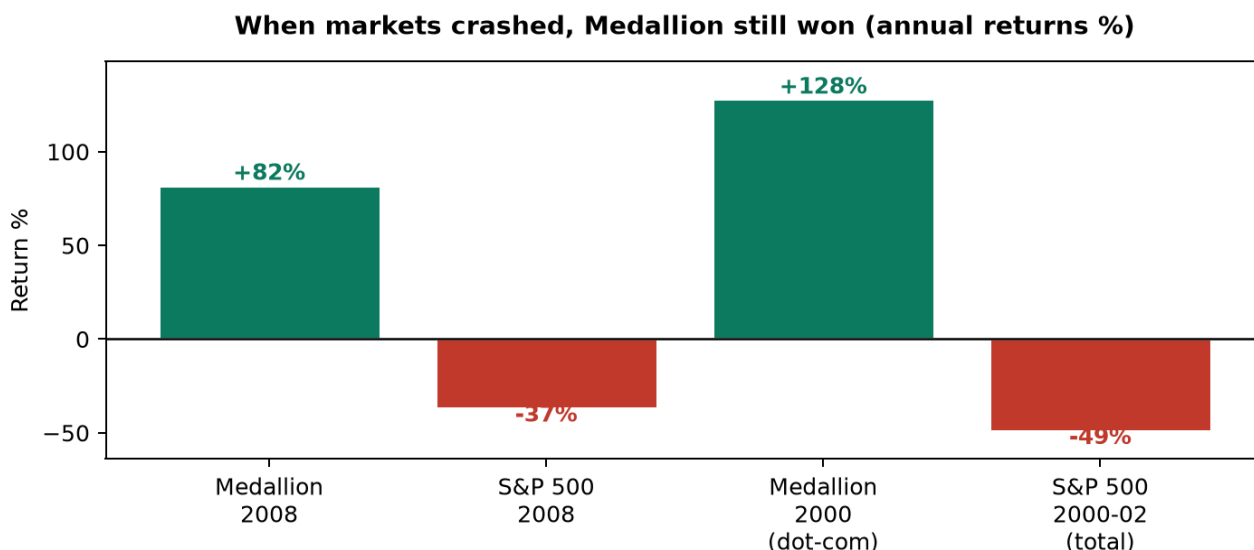


Figure 2 — Crisis years: Medallion was up big when markets crashed. (Net figures widely reported; gross was even higher.)

Plain-English takeaway: Imagine two snowballs rolling for 30 years. The S&P; snowball grows 10% a year. The Medallion snowball grows 40% a year *after* paying huge fees. Compounding turns that gap into a **mountain vs a pebble**.

3 • Who is Jim Simons? The math teacher who beat everyone

Jim Simons (1938–2024) wasn't a Wall Street banker. He was a **math professor, code-breaker and geometry prize-winner** who believed markets — like codes and shapes — hide **patterns you can find with math**. In 1982 he founded Renaissance Technologies and hired **mathematicians, physicists and computer scientists** instead of traders — including speech-recognition experts from IBM who had taught computers to understand human language.

Their big insight: **don't hire people with market opinions — hire people who find patterns in data**. Early years were messy (the fund even lost money in 1989). But by the 1990s the system clicked — and then it compounded at ~40% net for decades. Simons stepped back around 2009–2010, yet returns **continued and even improved gross (75–80%)** — proof the **system**, not one man, was the edge. The story is told in Gregory Zuckerman's book *The Man Who Solved the Market*.

"What you're really modeling is human behavior. Humans are most predictable in times of high stress — they act instinctively and panic." — Kresimir Penavic, Renaissance

"There's no data like more data." — Jim Simons' famous rule: collect everything, back to the 1800s if possible, and let the computer find what repeats.

4 • The strategy in ONE sentence (then in plain English)

ONE SENTENCE: Medallion uses computers to find **small, statistically proven price patterns** across hundreds of markets and trades them **automatically, thousands of times**, hedged and risk-controlled, holding **minutes to days**.

The casino analogy (the whole strategy in 20 seconds): A roulette wheel pays the casino only ~2.7% per spin — tiny. But the casino takes that tiny edge **thousands of times a day, every day, for years** — and gets rich

with near-certainty. Medallion is the casino: its edge per trade is **tiny** (often well under 1%), but it trades **massive volume, diversified everywhere, with strict limits** — so the edge compounds into ~66% a year before fees. You don't need to be right big. You need to be **right slightly, very often**.

You don't need a BIG edge — you need a SMALL edge, repeated thousands of times

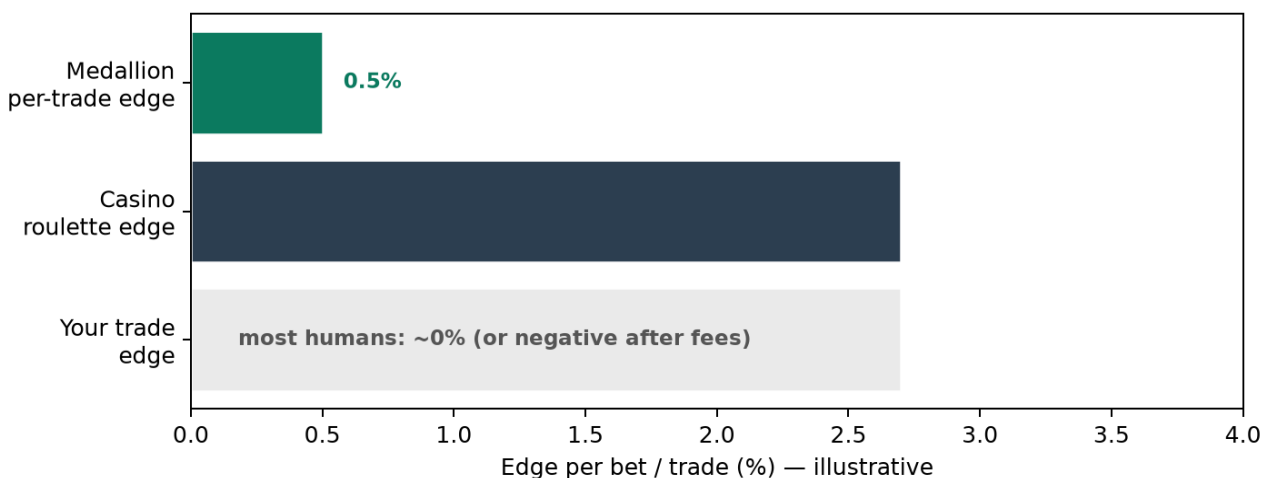


Figure 3 — Illustrative: casinos and Medallion both win with SMALL edges repeated at SCALE. Most human traders have no edge at all after costs.

5 • The 7 secrets — explained like you're 12

Nobody outside Renaissance knows the exact code — it's one of the best-kept secrets in finance. But from Zuckerman's book, employee quotes and academic work, the **framework** is well understood. Here are the 7 pillars, each with a simple analogy.

SECRET 1 — Don't ask WHY, ask WHAT happens next

Forget the news story. Medallion doesn't care *why* a stock moved (CEO, earnings, wars). It only asks: "**When this pattern happened 10,000 times before, what happened next?**" If the answer repeats reliably, it bets on it.

Analogy: Weather app vs. weather philosopher: you don't need to know *why* it rains — you just need to know that **dark clouds + wind = rain 80% of the time**, so you carry an umbrella.

Your move: So what do I do? Trade **rules and backtests**, not opinions. If you can't show a pattern worked 100+ times in the past, don't trade it.

SECRET 2 — Tiny edges × thousands of trades = fortune

Each Medallion trade aims for a **tiny profit**. Win rate is often only ~50–55% — barely better than a coin flip. But with **hundreds of thousands of trades**, that sliver becomes enormous. Short holding periods (minutes to days) also mean **huge sample sizes**, so they can prove an edge is real, not luck.

Analogy: The shovel seller in a gold rush: he doesn't find one giant nugget — he sells **10,000 small shovels** and keeps a cut of each.

Your move: So what do I do? Stop hunting one 10× trade. Build a system with a **small, repeatable edge** you can take 100s of times with low costs.

SECRET 3 — Prices snap back like rubber bands (mean reversion — the CORE engine)

For over a decade, Medallion's bedrock was **reversion to the mean**: when a price stretches too far from its usual level or from its relatives, it tends to **snap back**. Classic example: futures that open far below yesterday's close tend to bounce up that day — and vice versa. One insider summed it up: "**We make money from the reactions people have to price moves**" — i.e., from panic and overreaction.

Analogy: A stretched rubber band: pull it too far and **let go — it flies back**. Medallion bets on the snap-back, with computers measuring exactly how far is "too far."

Your move: So what do I do? Learn basic mean-reversion tools (Bollinger Bands, RSI extremes, pairs). Only fade extremes **with a stop** — rubber bands can snap *and keep going*.

SECRET 4 — Sometimes, ride the wave (trend / momentum)

Not everything snaps back — sometimes winners keep winning for a while. Medallion also runs **trend models** (e.g., stocks that rallied last week tend to drift further short-term). They blend reversion + trend so the system makes money in **both choppy and trending** markets. When a style stops working (their momentum model bled in the dot-com bust), they **switch it off automatically**.

Analogy: Surfing: most waves fizzle (fade them), but when a big one builds, you **hop on and ride** — then jump off before it crashes.

Your move: So what do I do? Don't marry one style. Have **one trend rule + one range rule**, and track which regime you're in.

SECRET 5 — Bet on everything, a little bit (extreme diversification)

Medallion trades **hundreds of markets** — stocks, futures, currencies, commodities worldwide — long AND short. Each bet is small and **uncorrelated**, so no single trade can hurt. Net market exposure stays near zero (**market-neutral**): they profit from **relative** moves, not from guessing whether the market goes up or down. That's why they made money in 2008 while holding almost no directional risk.

Analogy: 1,000 lottery tickets in 1,000 different draws vs. all your money on one ticket. One loss means nothing; the **average** is what pays you.

Your move: So what do I do? Never go all-in. Risk **1–2% per trade**, spread across a few uncorrelated ideas, and use hedges (or simply smaller size).

SECRET 6 — Let the robot trade. Never your feelings (100% systematic)

Every Medallion trade is dictated by the model — **zero human override**. Humans panic, get greedy, and "feel" the market is different this time. The computer doesn't. They even **hide their own trades** (scaling in/out at random times) so competitors can't copy them. Rule: if you override the system, you can't backtest it — so you no longer *have* a system.

Analogy: Autopilot on a plane: in a storm, the panicking passenger would yank the controls and crash. The **autopilot just follows its instruments** and lands safely.

Your move: So what do I do? Write your rules on one page. **If it's not in the rules, you don't trade it.** Journal every override — you'll quickly see they cost you money.

SECRET 7 — Borrow carefully + automatic brakes (leverage + risk engine)

Because each edge is small but high-probability and diversified, Medallion uses **big leverage (~12–20×)** to magnify it — like a magnifying glass on sunlight. Crucially, leverage sits on top of an **automatic risk engine**: when volatility spikes or a model misbehaves, positions **shrink automatically**. Contrast LTCM (blew up): *"LTCM believed its models were truth. We never believed ours reflected reality — just some aspects of it."*

Analogy: A race car with amazing brakes: the engine (leverage) makes it fast, but the **brakes (risk controls)** are why the driver survives. Amateurs add engine with no brakes — and crash.

Your move: So what do I do? Beginners: **NO leverage** until profitable for 6–12 months. Then tiny leverage, always with a **max daily loss + per-trade stop** that cuts you off automatically.

6 • A real example: the Coca-Cola vs Pepsi trade

The cleanest illustration of Medallion-style **statistical arbitrage / pairs trading** — no jargon needed:

- **Step 1 — Find two twins.** Coca-Cola and Pepsi almost always move together (same industry, same customers). The computer studies 20 years of data and learns their normal gap.
- **Step 2 — Wait for a weird stretch.** One day Coke jumps +3% on hype while Pepsi sits flat. History says: **9 times out of 10, this gap closes within hours/days.**
- **Step 3 — Bet on the snap-back (hedged).** The system **shorts the expensive one (Coke) + buys the cheap one (Pepsi)** in matched sizes. If the whole market crashes, both fall together — the *gap* is what matters, not the direction.
- **Step 4 — Exit when normal returns.** Gap closes → both legs closed for a **small profit**. Gap widens past a limit → **auto stop-out**, tiny loss. No debating, no hoping.
- **Step 5 — Repeat 10,000×.** Coke/Pepsi is one pair. Medallion runs this logic on **thousands of pairs and patterns across the globe, simultaneously** — that's the fortune.

Why this is genius: You're not predicting whether soda stocks go UP or DOWN (hard). You're predicting that **two twins will move back together** (much easier, statistically). Multiply by leverage + thousands of twins = Medallion.

7 • Why you can't just copy-paste it (5 barriers)

If it's so good, why doesn't everyone do it? Because Medallion's moat is enormous:

#	Barrier	What it means for you
1	Secret code + top scientists	90+ PhDs, decades of research, one giant secret model. You can't download it.

#	Barrier	What it means for you
2	Insane data + computing	Every tick, order-book depth, news sentiment — back to the 1800s. Costs millions.
3	Super-cheap trading costs	Tiny edges die if fees/slippage are high. Medallion executes better than any retail broker.
4	Bank leverage on tap	12–20× leverage at elite rates because their Sharpe ratio (~2–7) is absurd. You won't get those terms.
5	Capacity cap	Capped at ~\$10–15B and closed since 1993 — the edges <i>only</i> work at limited size. Scale kills them.

Bottom line: **you can't BE Medallion. But you can THINK like Medallion** — and that alone puts you ahead of 90% of traders. That's next.

8 • Your retail version: 5 rules you CAN copy

You don't need a PhD or a supercomputer. Steal the **principles**, shrink them to your size:

RULE 1 — Trade rules, not feelings (be the casino)

Write a 1-page plan: **when you enter, when you exit, how much you risk**. Backtest it on 100+ past trades (even by hand on TradingView). If it has no proven edge, don't fund it. Medallion's version: *"Don't override the computer."* Your version: **don't override the plan**.

RULE 2 — Small edge, many reps, low costs

Pick **ONE simple setup** (e.g., RSI pullback in an uptrend, or opening-range breakout) and trade only that, 100s of times. Keep costs tiny: liquid markets, limit orders, no overtrading. Track win rate + average win/loss — your "casino math."

RULE 3 — Mean-reversion starter (the Medallion flavor)

Example template (educational, not advice): on a daily uptrend (price above 200-day average), when RSI(2) drops below 10 (stretched rubber band), buy small with a stop below the recent low and exit at the 5-day average. Backtest first. Expect **many small wins + occasional small losses** — exactly the Medallion shape, minus leverage.

RULE 4 — Risk like a paranoid robot

Risk **1% per trade, max 3–5% per day**. If you hit the daily stop, **you're done — computer says no**. No averaging down, no "it'll come back." Size = Risk ÷ Stop distance. Beginners: **no leverage**. This single rule prevents 90% of blow-ups.

RULE 5 — Diversify + review weekly

Trade 2–4 uncorrelated ideas/markets instead of one. Every weekend, review: **did I follow rules? What's my edge this month? What should I switch OFF?** (Medallion kills dead models fast — you should kill dead setups fast too.)

Your one-page plan template: (1) Market + timeframe (2) Setup + entry trigger (3) Stop-loss (4) Take-profit (5) Position size (1% risk) (6) Max trades/day (7) When I STOP trading (daily loss / 3 losses in a row). **Print it. Sign it. Follow it like code.**

9 • How the other legends compare

Medallion is #1 on long-term documented returns. But other legends teach different lessons:

Legend / Strategy	Record	Style in simple words	Lesson
Medallion (Simons)	~39–40% net, 30+ yrs, 0 down yrs	Computer patterns, minutes–days, hedged	Systems + data beat gut feeling
Warren Buffett	~20% for 60 yrs	Buy great companies cheap, hold forever	Patience + compounding wins
Turtle Traders (Dennis)	\$1,600 → \$200M; trend rules	Ride big trends with breakouts	Simple rules + discipline work
George Soros	£1B in 1 day (1992)	Huge macro bets when odds skew	Bet big when you're certain
Paul Tudor Jones	+125% in 1987	Macro + charts + strict stops	Defense first: cut losses fast
Larry Williams	+11,376% in 1 yr (1987)	Short-term seasonal/commodity swings	Small accounts can sprint — then bank it
Dalio / Bridgewater	Largest hedge fund; All-Weather	Balanced across economies	Diversify across regimes

Notice the pattern: **every legend has strict rules + risk control**. None of them "YOLO trade on vibes." Style differs; **discipline doesn't**.

10 • Honest warnings (read this before you trade)

- **Past ≠ future.** Even Medallion's edges decay — they constantly research new ones and kill old ones. Any strategy you copy can stop working.
- **Leverage kills.** Medallion's 12–20× leverage works because of elite hedging, costs and auto-brakes. Retail leverage without those = fast account death. Start with zero leverage.
- **Costs eat small edges.** Spreads, commissions and slippage destroy high-frequency ideas at retail size. Trade liquid markets, use limits, trade less often than you want to.
- **Survivorship bias is real.** You hear about Larry Williams' +11,376% year — not the thousands who blew up trying. Trade money you can afford to lose.
- **No course can sell you Medallion.** Anyone claiming to sell "the Medallion strategy" is lying — the code has never left Renaissance. Learn principles, not promises.

THE GOLDEN RULE: If you can't explain your edge in one sentence, backtest it on 100+ trades, and state your max loss before you click — **don't click**.

11 • Your 30-day starter plan

Week	Do this	Done when...
Week 1	Read this guide twice. Pick ONE market + ONE setup. Open a demo account.	1-page plan written
Week 2	Backtest your setup on 100 past trades (TradingView replay). Log win% + avg win/loss.	You know your edge (or lack of it)
Week 3	Demo-trade the plan with 1% risk rules. No overrides. Journal every trade.	20+ demo trades, rules followed
Week 4	Review: keep, tweak or kill the setup. Only fund a tiny live account if demo was disciplined + profitable after costs.	Go / no-go decision, in writing

Recommended next reads: *The Man Who Solved the Market* (Zuckerman) for the story • Cornell's "Medallion Fund: The Ultimate Counterexample" for the math • *Market Wizards* (Schwager) for trader psychology • Your broker's demo + TradingView for practice.

12 • Sources & further reading

Key figures in this guide (66% gross / ~39–40% net, \$100 → \$398.7M, zero losing years, 2008 +82% net, 5-and-44 fees, 150k+ trades/day, 12–20× leverage) are the widely-reported numbers from the sources below. Medallion is private and doesn't publish audited statements — figures are reconstructions from reporting + academia, but they are consistent across sources.

[1] Cornell, B. — “Medallion Fund: The Ultimate Counterexample” (Cornell Capital Group). \$100 → \$398.7M (1988–2018), 63.3% CAGR gross, 66.1% mean, zero down years, beta ≈ −1.

[2] Zuckerman, G. — *The Man Who Solved the Market* (2019) + interviews: reversion core, 5-minute patterns, leverage via high Sharpe, auto risk cuts, “basket” trades.

[3] Quartr / TrendSpider / Acquired.fm deep-dives on Renaissance: 39.9–40% net CAGR 1988–2022, \$1,000 → ~\$90M net, IBM speech-team hires, Kelly sizing, capped fund.

[4] 247WallSt (2026) + IWP Finance case study: 5% + 44% fees, gross rising to 75–80% post-Simons, Senate/tax history, Sharpe > 2.

[5] Novel Investor notes on *The Man Who Solved the Market*: pair/reversion quotes, “we make money from reactions to price moves,” momentum switch-off in 2000.

[6] HyroTrader (2026, data roundup): trend-following 29–58% CAGR studies, mean-reversion 68–71% win rates, breakout +1,600% (2016–23) — context for Sections 5 & 9.

[7] Strike.Money / Plus500 GOAT lists: Buffett ~20%, Soros £1B day, Tudor +125.9% (1987), Williams +11,376% (1987), Lynch 29.2% — context for Section 9.

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